

EN401 — Handout 5 Results & Analysis Report

Commercial Thermal Generation Era (Gas Turbines & Diesel Generators)

1. Scenario Objective & LP Solution Metrics

- **Objective Value (NPV):** \$19,335.91 (-99.96% vs HO4)
- **Problem Scale:** 1,502 constraints, 960 variables, 8,640 non-zeros
- **Technologies Added:** PWRDSL, PWRNGS, PWRTRN, PWRDIST

2. Power Plant Capacity Additions

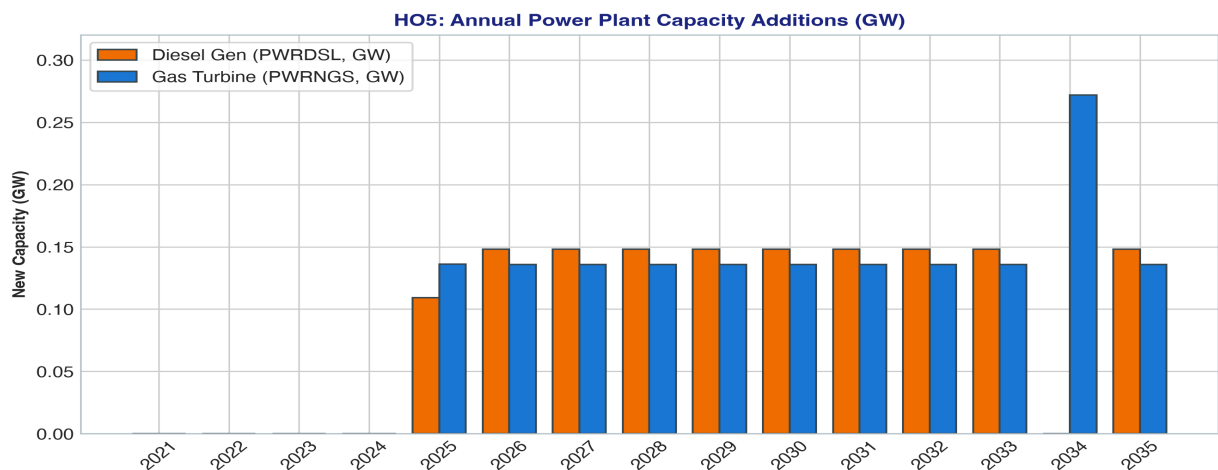


Figure: Annual capacity additions of gas turbines and diesel generators (GW).

3. System Cost Reduction (-99.96%)

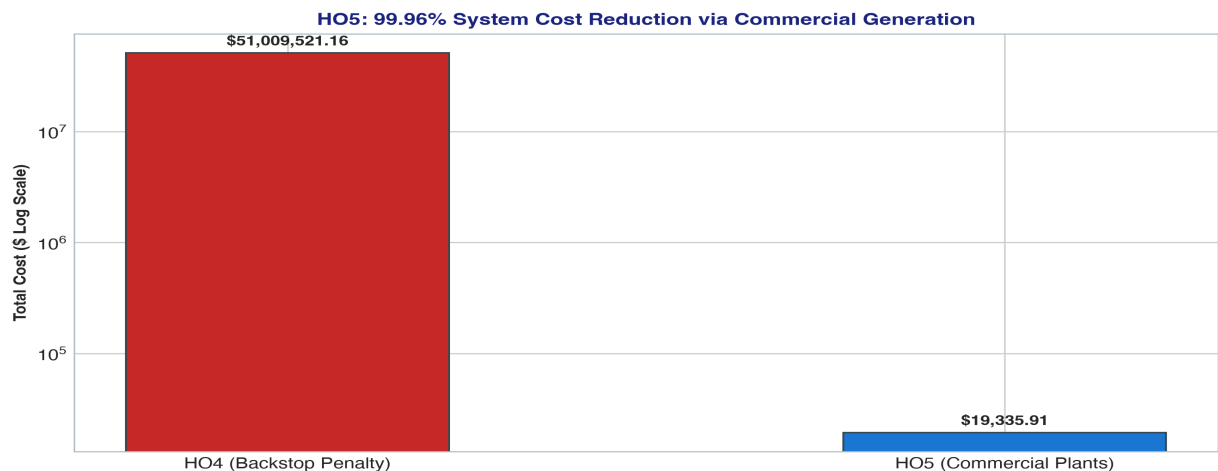


Figure: Displacement of backstop penalty drops system cost to \$19,335.91.

4. Engineering Takeaways

Gas turbines and diesel generators share the market: natural gas supplies low-cost baseload energy, while diesel generators meet diurnal peaks.